

Arnab Das

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CAREER OBJECTIVE

Computer Science graduate with a strong foundation in AI, Robotics and Digital logic, looking to move into Electrical and Computer Engineering to gain hands-on research experience and contribute to the development of intelligent and efficient systems

EDUCATION

2025 Bachelor of science in Computer Science at **BRAC University** (CGPA: 3.89/4.00)

RELEVANT COURSEWORK

CSE461: Introduction to Robotics

- Studied forward and inverse kinematics for robotic manipulators and industrial arm configurations
- Practiced trajectory planning techniques including joint space, linear, and non-linear path planning
- Learned PID control theory, block diagram analysis, and open/closed loop system behavior
- Studied robot navigation concepts including path planning, localization, mapping, and occupancy grids
- Applied machine learning and neural network concepts to robotic perception and decision-making tasks
- Learned convolutional neural networks (CNNs) for computer vision applications in robotics

CSE422: Artificial Intelligence

- Studied AI foundations including the Turing test, rational versus human behavior, and AI agents
- Learned informed and local search including state space representation, heuristic formation, greedy best-first search, A* search, heuristic admissibility and consistency, hill climbing, simulated annealing, and gradient descent
- Solved problems using genetic algorithms for the n-queens, knapsack, and traveling salesman problems
- Covered two-player game trees including minimax and alpha-beta pruning
- Studied probability theory including joint probability distributions and independence, and covered naive Bayes classification using Bayes theorem
- Learned supervised learning through linear and logistic regression to predict outcomes, trained perceptron and neural network models using gradient descent and backpropagation, and built decision trees for classification using the ID3 algorithm, entropy, and information gain

CSE330: Numerical Methods

- Studied floating-point arithmetic including fixed-point and floating-point number representation, significant figures, rounding error, and loss of significance
- Learned polynomial interpolation methods including Taylor series, Lagrange and Newton divided-difference forms, interpolation error, and convergence with Chebyshev nodes
- Covered numerical differentiation and root-finding methods including finite differences, Richardson extrapolation, interval bisection, fixed point iteration, Newton's method, and quasi-Newton methods
- Solved systems of linear equations using Gaussian elimination, LU decomposition, and pivoting, and studied vector and matrix norms and conditioning
- Studied least-squares approximation including orthogonality, discrete and continuous least squares, QR decomposition, and orthogonal polynomials
- Learned numerical integration techniques including Newton-Cotes formulae, composite rules, and Gaussian quadrature

CSE260: Digital Logic Design

- Learned number systems and binary arithmetic, and studied Boolean algebra and simplification techniques
- Used Karnaugh map and tabulation methods to minimize combinational circuits and calculate SOP and POS expressions
- Designed and analyzed combinational circuits including adders, subtractors, comparators, decoders, encoders, and multiplexers
- Designed and analyzed sequential circuits including flip-flops, counters, registers, and memory elements
- Built and troubleshot combinational and sequential circuits through weekly lab work and a design project

STA301: Modern Probability Theory & Stochastic Processes

- Reviewed core probability theory including conditional probability, independence, random variables, joint/marginal/conditional distributions, and expectation and variance
- Studied discrete distributions (binomial, Poisson, hypergeometric, geometric, negative binomial) and continuous distributions (uniform, exponential, gamma, normal), along with sampling distributions and the Central Limit Theorem
- Learned the fundamentals of stochastic processes including state space, the Markov property, transition probability matrices, and the Chapman-Kolmogorov equation
- Analyzed Markov chains covering limiting behavior, classification of states, irreducibility, and Hidden Markov Models
- Studied applied stochastic models including random walks, the Gambler's Ruin problem, Poisson processes, and birth-death and queueing processes

STA201: Elements of Statistics & Probability

- Learned data summarization and descriptive statistics including graphical presentation, measures of central tendency, dispersion, box plots, and outlier detection
- Studied correlation and simple/multiple linear regression analysis
- Covered foundational probability concepts including sample space, events, addition and multiplication rules, conditional probability, independence, and Bayes' theorem
- Explored random variables and probability distributions including expectation, variance, joint distributions, and binomial, geometric, Poisson, normal, and exponential distributions
- Learned statistical hypothesis testing including one-sample and two-sample z-tests and t-tests for population means

MAT216: Linear Algebra & Fourier Analysis

- Learned linear systems including row reduction, echelon forms, vector equations, and linear independence
- Studied matrix algebra including matrix operations, matrix inverses, determinants, and applications to computer graphics
- Covered vector spaces and subspaces including null, column, and row spaces, basis, and rank
- Learned eigenvalues, eigenvectors, and diagonalization, and studied orthogonality using inner products, projections, and the Gram-Schmidt process
- Studied methods for solving boundary value problems and their applications
- Studied Fourier series, orthogonal functions, and Fourier sine and cosine transforms

MAT215: Complex Variables & Laplace Transformations

- Learned complex numbers and functions including algebraic and polar forms, Euler's formula, De Moivre's theorem, and exponential, logarithmic, and trigonometric functions of a complex variable
- Studied analytic functions including Cauchy-Riemann equations, harmonic functions, and conformal mappings
- Covered complex integration including Cauchy's Integral Theorem, Cauchy's Integral Formula, and contour integration
- Studied series expansions and singularities including Taylor and Laurent series, classification of singular points, and poles
- Solved problems using residue calculus to evaluate real integrals and applied Laplace transforms to solve ordinary differential equations

PROJECTS

AetherGuard: A Private Password Manager

[Link to Project](#)

Built a full-stack password manager using Python and Flask; implemented AES-256 encryption for stored credentials and Argon2 password hashing for user authentication, with full CRUD operations, random password generation, account management features, and a dark-themed glassmorphism web interface backed by a local SQLite database.

Demo Stock Exchange Interface

[Link to Project](#)

Built a Demo Stock Exchange desktop app with Python, CustomTkinter, and SQLite, featuring user authentication (login/signup with NID and phone validation), wallet management, portfolio tracking, stock trading, and a responsive UI with dark/light mode toggle.

Air Quality Prediction: ML Classification Project

[Link to Project](#)

Built an air quality prediction system on the UCI Air Quality dataset using Decision Tree, Random Forest, and KNN classifiers; engineered a CO-level target variable from CO_{GT} readings with full preprocessing pipeline (missing value imputation, column pruning, label encoding); achieved 99.68% test accuracy with Random Forest; conducted EDA with distribution plots, correlation analysis, and violin charts to derive environmental insights.

Network Intrusion Detection System

[Link to Project](#)

Built a binary network intrusion detection system using the KDD Cup 1999 dataset; applied MinMax normalization, PCA dimensionality reduction, and categorical encoding as a preprocessing pipeline; trained a deep neural network (Keras Sequential) with ReLU and Sigmoid activations to classify network traffic as intrusion or non-intrusion.

AWARDS AND CERTIFICATIONS

Vice Chancellor's List Appeared on the Vice Chancellor's List three times	Check Transcript
Career Essentials in Software Development Issued by Microsoft	View Certificate 2024
Career Essentials in Generative AI Issued by Microsoft	View Certificate 2024
Champion – Creative Talent Search Competition Organized by Government of the People's Republic of Bangladesh	View Certificate 2017

SKILLS

Programming Languages	Python, C, C++, JavaScript
Systems & Scripting	Bash Shell, Linux
Database	PostgreSQL
Documentation & Typesetting	Markdown, L ^A T _E X
Web Technologies	HTML, CSS, JavaScript
Tools & Technologies	Git